Quiz : Thurs. 3:40-4:00

Agenda

Finish LDA

GDA

Naive Bayes - activity

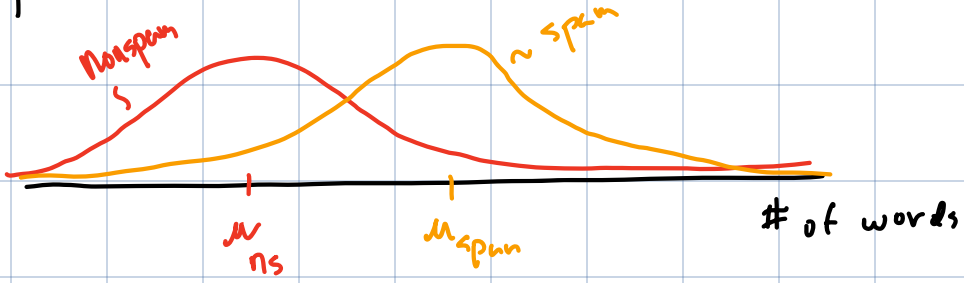
1. LDA

1.1 Recap

Assume $X \mid Y = k \sim N(\mu_k, \Sigma)$ ← Gaussian

↑ same for all classes k

$p = 1$, $K = 2$



$$p_{spam}(x) = \frac{\pi_{spam} f_{spam}(x)}{\pi_{spam} f_{spam}(x) + \pi_{ns} f_{ns}(x)}$$

← highest numerator wins

$$f_{spam}(x) = N(\mu_{spam}, \sigma^2) \qquad f_{ns}(x) = N(\mu_{ns}, \sigma^2)$$

share σ^2

1.2 Generally (k) $p=1$, $K=2$

$$f_k(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma^2}\right)$$

$$\log\left[\pi_k f_k(x)\right] = \log \pi_k + \log f_k(x)$$

don't depend on k

$$= \log \pi_k - \frac{1}{2}\log 2\pi - \log\sigma - \frac{(x-\mu_k)^2}{2\sigma^2}$$

$$\delta_k(x) = x\left(\frac{\mu_k}{\sigma^2}\right) - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k$$

Discriminant function

where: $\hat{\mu}_k = \frac{1}{n_k} \sum_{i: y_i = k} x_i$

$$\hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i: y_i = k} (x_i - \hat{\mu}_k)^2$$

← wt avg of sample variance for each class

$$\hat{\pi}_k = \frac{n_k}{n}$$

1.3 $p > 1$ $\quad X \mid Y = k \sim N(\mu_k, \Sigma)$

μ_k is $p \times 1$

Σ is $p \times p$ same for all k

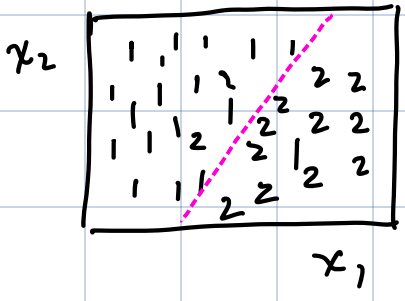
$$f_k(x) = \left(\frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}}\right) \exp\left[-\frac{1}{2}\underset{p\times 1}{(x-\mu_k)^T} \underset{p\times p}{\Sigma^{-1}} \underset{p\times 1}{(x-\mu_k)}\right]$$

$$\Rightarrow \delta_k(x) = \underset{1\times p}{x^T}\ \underset{p\times p}{\Sigma^{-1}}\ \underset{p\times 1}{\mu_k} - \frac{1}{2}\underset{1\times p}{\mu_k^T}\ \underset{p\times p}{\Sigma^{-1}}\ \underset{p\times 1}{\mu_k} + \log \pi_k$$

$$\hat{\Sigma} = \frac{1}{n-K}\sum_{k=1}^{K}\sum_{i:y_i=k}(x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$$

$(x_i - \hat{\mu}_k)$: $p\times 1$; $(x_i - \hat{\mu}_k)^T$: $1\times p$; product: $p\times p$

Notice $\delta_k(x)$ is linear



Boundary: $\delta_1(x) = \delta_2(x)$ $(K=2)$

$\delta_1(x) > \delta_2(x)$ classify as $Y=1$

else $Y=2$

2. Quadratic Discriminant Analysis (QDA)

2.1 One difference

$$X \mid Y=k \sim N(\mu_k, \Sigma_k)$$

↑ depends on k

$$\Rightarrow \delta_k(x) = -\frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(x-\mu_k)^T\Sigma_k^{-1}(x-\mu_k) + \log\pi_k$$

Notice: $x^T\Sigma_k^{-1}x$ is quadratic in x

2.2 Tradeoffs

			$p=2, K=2$
LDA	$p(p+1)/2$	covariances	3
QDA	$Kp(p+1)/2$	covariance	6

QDA requires $n_k > p$ or else $\hat{\Sigma}_k$ is singular

QDA more flexible, higher variance, lower bias

Use QDA if n is large or Σ is different for each k

3. Naive Bayes

3.1 Naive assumption: predictors are independent!

$$f_k(x) = f_{k1}(x_1) f_{k2}(x_2) \cdots f_{kp}(x_p)$$

$$p_k(x) \propto \pi_k f_{k1}(x_1) \cdots f_{kp}(x_p)$$

$f_k(x)$ no longer multivariate Normal

each predictor gets its own f_{kj}.

We can mix quant. & qual. predictors.

3.2 Example (binary predictors)

$n = 10$

$K = 2$ (spam, ns)

$p = 3$

Email	Class (Y)	X_1: free	X_2: link	X_3: contact
1	S	1	1	0
2	S	1	1	0
3	S	1	0	1
4	S	0	1	0
5	NS	1	0	1
6	NS	0	1	1
7	NS	0	1	1
8	NS	0	0	1
9	NS	0	1	0
10	NS	0	0	0

STEP 1: Priors ($\hat{\pi}_k$)

$\hat{\pi}_{spam} = 4/10$ $\hat{\pi}_{ns} = 6/10$

STEP 2: $\hat{f}_k(x) = \hat{P}(X = x \mid Y = k)$

	$P(\text{free}=T \mid k)$	$P(\text{link}=T \mid k)$	$P(\text{contact}=T \mid k)$
SPAM	3/4	3/4	1/4
NS	1/6	3/6	4/6

STEP 3: Classify new email with "free", a link, not in contacts. $x = (1, 1, 0)'$

$$\hat{\pi}_k \hat{f}_{k1}(x_1) \hat{f}_{k2}(x_2) \hat{f}_{k3}(x_3)$$

$$\text{SPAM} = 4/10 \times 3/4 \times 3/4 \times 3/4 = .169 \leftarrow \text{classify as SPAM}$$

$$\text{NS} = 6/10 \times 1/6 \times 3/6 \times 2/6 = .02$$

STEP 4: (optional)

$$\hat{p}_{\text{spam}}(x) = \frac{.169}{.169 + .02} \approx .91$$